

Foreclosure Deeds That Look Like Sales: Legal-Rule Amounts in Public Records and House-Price Indices

Research Note

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Abstract

Public sales files can carry legal consideration from lender acquisitions as if it were a negotiated price. In New York City, party-based classification flags 16,655 potentially affected deeds among 1.47 million records from 2003 to 2025. About 62% survive conventional screens, and 94-95% of screened lender acquisitions recorded since 2016 match the public sales file at the same amount. Removing pairs with flagged endpoints changes a house repeat-sales index by 3.69 log points in 2010Q3 (simultaneous 95% band: 1.59 to 5.78). A narrower definition gives 2.63 points. The estimated peak-to-trough change is small and its interval includes zero. A hedonic comparison also shows an excess around 2010, but lacks the later repeat-sales shortfall, consistent with the role of pairing. The released classifier supplies a reproducible screen; its accuracy has not been established against independent human document review. The contribution is a measurement application for researchers using public records without vendor transaction flags.

Keywords: administrative data; house-price index; repeat sales; foreclosure; record linkage

1. Introduction

Studies of house prices drop transfers that are not arm's-length sales, such as gifts, transfers between relatives and repossessions by lenders. Commercial data vendors flag transaction types for this purpose. Kermani & Wong (2026), for example, keep only the arm's-length transactions in vendor deed data, which excludes deeds in lieu of foreclosure, and the S&P Cotality Case–Shiller methodology excludes repossessions by lenders that the deed record identifies (S&P Dow Jones Indices, 2025). Researchers who work from free public files have only what the record shows, and the screens they apply are designed for gifts and nominal transfers.

This note documents a class of transfers that those screens miss in New York City. When a lender takes title through foreclosure or a deed in lieu, transfer-tax law can base the recorded consideration on a bid, a judgment or the discharged debt. That amount usually falls within the range of market prices, so it passes price screens, and the city's free sales file carries it with no transfer-type flag. Paired with the lender's later resale, it enters a repeat-sales index as if it were a price. The contribution is to quantify this measurement problem in a public registry and supply a reproducible screening rule. The need to exclude lender repossessions is established. Here, the amounts survive conventional filters and enter the public sales file, while their inclusion changes the timing of a repeat-sales index. A hedonic comparison tests whether the same pattern appears without repeat-sale pairing. Removing flagged observations changes the index's transaction population; the screened index is not a verified constant-quality counterfactual. The full working paper, with the classifier, a document review and a replication package, is Loschi (2026).

The closest work studies distressed sales as prices: foreclosure discounts (Campbell et al., 2011; Clauretje & Daneshvary, 2009; Conklin et al., 2023; Pennington-Cross, 2006) and the sensitivity of price indices to

including distressed transactions (Doerner & Leventis, 2015). Closer still is evidence that a recorded price can differ from the negotiated one for an institutional reason (Ben-David, 2011), and that crisis-era repeat-sales indices are biased when the quality of the paired homes changes (Nowak & Smith, 2020). The issue here comes before these: for some deeds the registry amount is not a price at all, and a researcher who works from public records inherits it unless the deed is classified by its parties.

2. Legal-rule amounts and how to find them

New York Tax Law §1401(d) and New York City Administrative Code §11-2101(9) include “the cancellation or discharge of an indebtedness or obligation” in consideration. For a lender acquisition at auction, the July 2019 state Schedule E starts from the higher of the foreclosure judgment and bid, adds continuing liens and, for recourse debt, applies a fair-market-value cap. For a deed in lieu it starts from unpaid secured debt plus continuing liens, with a recourse cap and any further payment. The city calculation has its own rules. The July 2019 form is not evidence that every earlier filing followed an identical formula (New York State Department of Taxation and Finance, 2019). These provisions identify possible legal bases; the name rule does not establish the exact basis of each recorded amount. When no third party outbids the plaintiff, the lender or its designee takes title, then sells the property as real-estate-owned (REO). In the data, the median interval between the lender’s acquisition and the next sale of the parcel is 498 days.

The Automated City Register Information System (ACRIS) records these conveyances as ordinary deeds; the deed type does not distinguish them. The parties do. A deed is classed as carrying a legal-rule amount when the grantee is a lender (a bank, servicer, federal agency or securitization vehicle, identified by institutional phrases rather than single words that can be surnames) and the conveyance is a referee’s deed, another transfer to a lender or a transfer between lenders. Lender resales are retained as potential market sales, separately from lender acquisitions. The rule labels 16,655 of 1.47 million deed-family documents recorded in Manhattan, the Bronx, Brooklyn and Queens from 2003 to 2025. They are 1.1% of deeds overall, but 3.7% of priced deeds in 2008 and 4.2% in 2009.

All inputs are public. The deed records, their parties, parcels, remarks and cross-references come from the ACRIS tables on NYC Open Data, retrieved month by month with each chunk checked against an exact row count. Sale prices come from the Department of Finance sales files. The hedonic attributes in Section 4 come from the Department of Finance annual sales files. The retrieval scripts, manifests and an archived snapshot are in the replication package (Loschi, 2026).

Three diagnostics describe how the flagged amounts differ from ordinary prices. They do not establish market status on their own. First, negotiated prices tend to be round, while debt balances need not be. Among screened sales that enter a repeat pair, 69.9% of legal-rule amounts carry non-zero cents, against 6.3% of ordinary prices, and 17.5% are multiples of \$1,000, against 81.2%. Second, the resale path is unusual: pairs whose first sale is a legal-rule deed fell in nominal terms 73.9% of the time, against 18.2% of pairs with no lender on either side. Third, against external benchmarks the amounts look like a price for the property as it was before the default. Relative to the parcel’s previous ordinary sale carried forward with the index, the median gap is +1.4 log points for referee’s deeds and +1.3 for other lender acquisitions. Relative to the parcel’s next sale, which is the lender’s resale 94% of the time, they sit 34.2 and 23.9 points above. Part of that gap is a forced-sale discount (Campbell et al., 2011), which the registry cannot separate from the recording rule.

The usual screens keep most of these deeds (Table 1). A positive amount and a \$10,000 minimum remove a third of them, mostly lender-to-lender transfers recorded at nominal amounts; full-interest, single-parcel and residential-class filters remove almost none. At the end, 62.2% of legal-rule deeds survive, against 51.6% of ordinary deeds, and 54.6% enter a screened repeat-sales pair, against 27.1%. The screens are designed for gifts and nominal transfers, and a debt balance is neither.

Table 1. Deeds surviving cumulative screens (deed-family documents, four boroughs, 2003–2025)

Screen (cumulative)	Legal-rule	Share	Ordinary	Share
All deed-family documents	16,655	100.0%	1,430,816	100.0%
Document type DEED	16,068	96.5%	1,354,845	94.7%
Amount > 0	14,424	86.6%	872,576	61.0%
Amount ≥ \$10,000	10,838	65.1%	846,528	59.2%
Full interest conveyed	10,832	65.0%	840,379	58.7%
One parcel, no easement, air rights or partial lot	10,667	64.0%	795,142	55.6%
Residential class	10,361	62.2%	738,334	51.6%
In a screened repeat-sales pair	9,091	54.6%	387,872	27.1%

Source: Loschi (2026), Table 3. The two columns describe classified subsets, not the entire deed universe: the ordinary column excludes lender resales and third-party auction purchases. The Department of Finance sales file, a standard source for New York City price series (e.g., Schwartz et al., 2003; Ellen et al., 2007), carries the same amounts: of screened legal-rule deeds recorded in 2016–25, 93.8% of referee’s deeds to lenders and 95.3% of other lender acquisitions appear in it for the same parcel within seven days at the same amount.

3. The effect on a repeat-sales index

The index is the quarterly estimator of Bailey et al. (1963), fitted to 140,119 pairs of consecutive screened sales of 1–3 family houses, 2003–2025. Pairs require a holding period of at least 180 days and an annualized log change no larger than one in absolute value. The correction drops every pair with a legal-rule endpoint (12,483 pairs) and keeps the rest unchanged.

The mechanism follows from the pairs. If a legal-rule amount L exceeds what the property would fetch, the pair from the previous market purchase A to L overstates appreciation into the take-title quarter, and the pair from L to the lender’s resale B overstates depreciation into the resale quarter. The index should therefore be too high when lenders take title and too low when they resell.

Table 2 shows the predicted pattern in the index comparison. The raw index lies above the corrected one by up to 3.69 log points in 2010Q3. A simultaneous 95% band for the whole path, from a paired parcel bootstrap, runs from 1.59 to 5.78 at that quarter. The raw index is 1.07 points higher on average over 2008–10 and 2.28 points lower over 2011–12, both with intervals that exclude zero. The estimated peak-to-trough contrast is –0.64 log points, with an interval from –3.03 to +1.25. The point estimate changes the timing more than the cumulative decline; the interval does not establish that the latter is unchanged. The difference is concentrated where lender acquisitions are: for Bronx houses it reaches 10.92 log points in 2010Q3 (interval 6.64–14.70), while for condominiums it is negligible.

Three checks address the obvious objections. A narrower definition that drops only referee’s deeds to lenders with non-zero cents and no securitization-type grantee, about a third of the flagged set, still moves the 2010Q3 index by 2.63 log points. A placebo that deletes random ordinary pairs with the same borough and timing moves it by at most 3.41 points in 200 draws for the published definition, so the observed 3.69 clears it narrowly, and by at most 1.89 for the narrowest definition, which 2.63 clears by more. Five explicit assignments for unresolved reviewed deeds preserve the 2011–12 shortfall. They do not bound classification error among unreviewed deeds. A further check restricts both endpoints to houses: the 2010Q3 differences are 3.61 and 2.62 points. Under the narrowest definition, one of 999 degree-matched placebo draws exceeds 2.62. Placebo margins depend on the deletion design and are descriptive references, not calibrated p-values.

Table 2. Raw minus corrected index, 1–3 family houses, log points

	Published definition	Narrowest definition
<i>Repeat sales (all pairs)</i>		
Difference in 2010Q3 [simultaneous 95% band]	+3.69 [+1.59, +5.78]	+2.63 [+0.88, +4.37]
Mean difference 2008–10 [95% CI]	+1.07 [+0.55, +1.59]	+1.05 [+0.83, +1.24]
Mean difference 2011–12 [95% CI]	–2.28 [–2.76, –1.72]	–1.09 [–1.27, –0.91]
Largest difference in aligned placebo (maximum of 200)	3.41	1.89
Peak-to-trough difference [95% CI]	–0.64 [–3.03, +1.25]	+0.63 [–0.77, +0.98]
<i>Hedonic index</i>		
Difference in 2010Q3 [simultaneous 95% band]	+2.07 [+1.27, +2.87]	+1.55 [+0.94, +2.16]
Mean difference 2011–12 [95% CI]	+0.14 [+0.02, +0.27]	+0.10 [+0.06, +0.13]
Repeat sales on the hedonic sample, mean 2011–12	–2.31	–1.03

Notes: Screened = pairs (repeat sales) or sales (hedonic) with a flagged deed dropped. Bands condition on the observed labels and do not include classification uncertainty. The narrowest definition keeps only referee’s deeds to lenders with non-zero cents and no securitization-type grantee. Intervals come from paired Poisson(1) parcel bootstraps with 300 draws. The aligned placebo deletes random ordinary pairs matched to the dropped pairs on borough, first-sale quarter and second-sale quarter. The hedonic bands are joint across both hedonic specifications, both definitions and all quarters. Source: Loschi (2026), Tables 4–6 and 8.

4. A hedonic check

A hedonic index uses screened sales without pairing successive transfers. Comparing it with repeat sales examines whether the later shortfall persists under a different estimator. An observation also affects fitted covariate coefficients and normalization, so its effect need not remain confined to its recording quarter. The

index regresses the log recorded amount of 435,486 screened sales of 1–3 family houses on quarter, ZIP-code and building-class indicators and structural attributes (floor and lot area, age, number of units), taken from the Department of Finance’s annual sales files.

In 2010Q3 the raw hedonic index lies 2.07 log points above the index without these deeds, with a simultaneous band from 1.27 to 2.87, and 1.55 above under the narrowest definition (Table 2 and Figure 1). The building descriptions in the 2003–2010 files date from May 2012, so they can postdate the sale; a model with only quarter, ZIP-code and class indicators gives 1.89 and 1.36. Over 2011–12, however, the hedonic difference averages only 0.14 log points. A repeat-sales index restricted to pairs with both endpoints in the hedonic sample keeps its shortfall (–2.31). This rules out the different attribute coverage as the sole explanation. The pattern is consistent with the influence of pairing a lender acquisition with its resale; it does not identify the market value of either transfer.

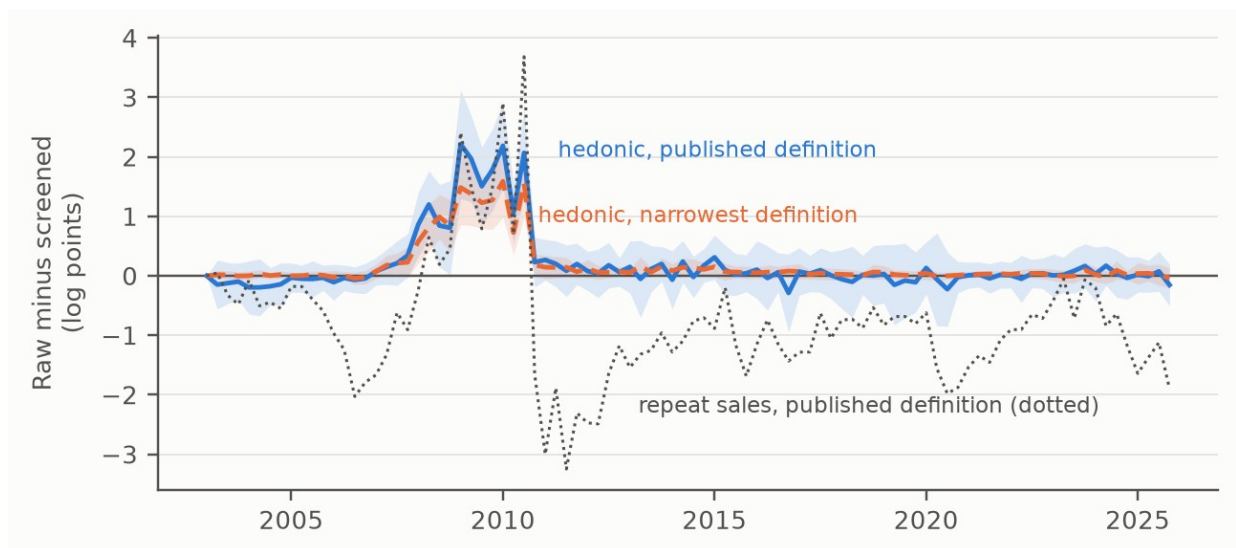


Figure 1. Raw minus screened index for 1–3 family houses, log points. Solid and dashed: hedonic index under the published and the narrowest definition, with 95% bands joint across both plotted definitions and all quarters. Table 2 uses wider bands that also include the class-and-location specification. Dotted: repeat-sales index, published definition.

5. How accurate is the classification?

To measure error rates, a stratified probability sample of 400 screened sales was drawn and each deed image read without its label. The review used AI in place of human readers: OpenAI’s ChatGPT read each image and cited the pages it relied on, and Anthropic’s Claude checked every reading against the city’s open data. The rates below therefore measure agreement with those readings, not accuracy against a verified record. Of the 400 sampled deeds, 316 receive a definite reading and 84 remain unresolved. Reported intervals condition on the resolved readings; excluding unresolved cases need not be innocuous. Among resolved deeds, the classifier’s precision is 0.993 (95% interval 0.978–1.000). Its recall for foreclosure-type transfers is 0.966 (0.934–0.998); the misses are mostly lender-side buyers whose names carry no institutional word. The classifier does not look for sales between relatives, which screens for nominal prices are meant to catch. The index calculations hold the labels fixed. The implemented name rule does not use consideration, but cents patterns helped identify lender subsidiaries during its development. The digit evidence is therefore a

diagnostic on the development data, not an independent validation. The narrowest definition explicitly uses cents and still gives the index pattern in Table 2.

6. Conclusion

Public transaction files can retain lender-acquisition amounts that conventional price screens do not remove. In New York City, excluding flagged endpoints changes the house repeat-sales index by 3.69 log points in 2010Q3, and by 2.63 under a narrower definition. The estimated cumulative peak-to-trough change is -0.64 points, with a 95% interval from -3.03 to 1.25. The result concerns the path of an index conditional on its screens and classifications.

The hedonic comparison supports an excess around lender acquisition without reproducing the later repeat-sales shortfall. It is consistent with the pairing mechanism, while leaving property condition and the market value of individual transfers unresolved. The practical step is to inspect the jurisdiction's consideration rules and join party records before treating every positive deed amount as a price. The released classifier makes that screen reproducible. Its labels and unresolved document cases should remain visible in sensitivity analysis (Loschi, 2026).

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Author Contributions

Pablo Loschi is the sole author and is responsible for the research design, data curation, analysis, software, writing and revision.

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Declaration of conflicting interest. The author founded HolaCasaClara, a free comparator of Argentine mortgage loans, which has no connection to New York housing records or to this study. The author declares no other conflicts of interest.

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Ethical considerations. The study analyses public administrative records without recruiting or interacting with participants. No ethics committee approval or waiver is claimed.

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Consent for publication. Not applicable. No identifiable document images or personal case narratives are reproduced in this note.

Data availability statement. All inputs are public New York City Open Data records, Department of Finance sales and valuation files and FRED series. Code, the classifier, the document-review sample and archived inputs are at <https://doi.org/10.5281/zenodo.22925383>.

Use of generative AI. The author used OpenAI's ChatGPT and Codex, Anthropic's Claude and Google's Gemini to draft and revise text and to write and check code. The document review in Section 5 used AI in place of human readers, as described there. The author reviewed and checked all output and takes full responsibility for the content.

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